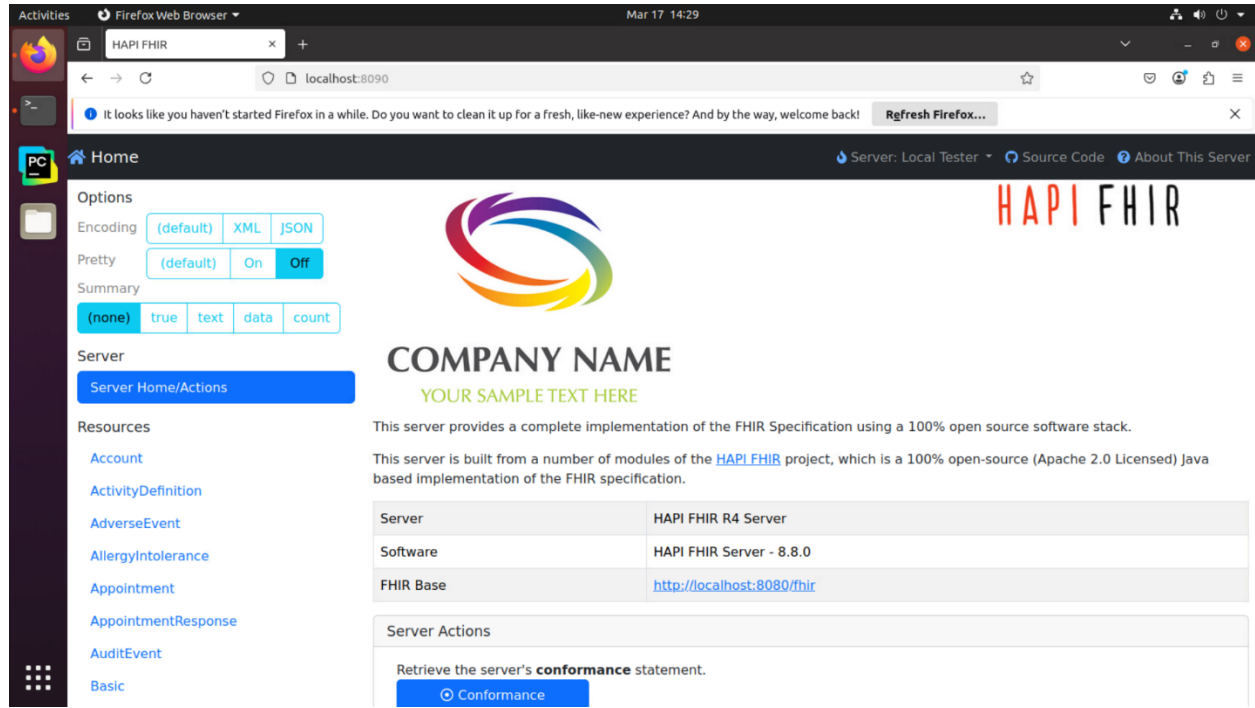


Architecture Development Part 4: Installation and Configuration of Hapi-FHIR Server

1. Hapi-FHIR Server UI



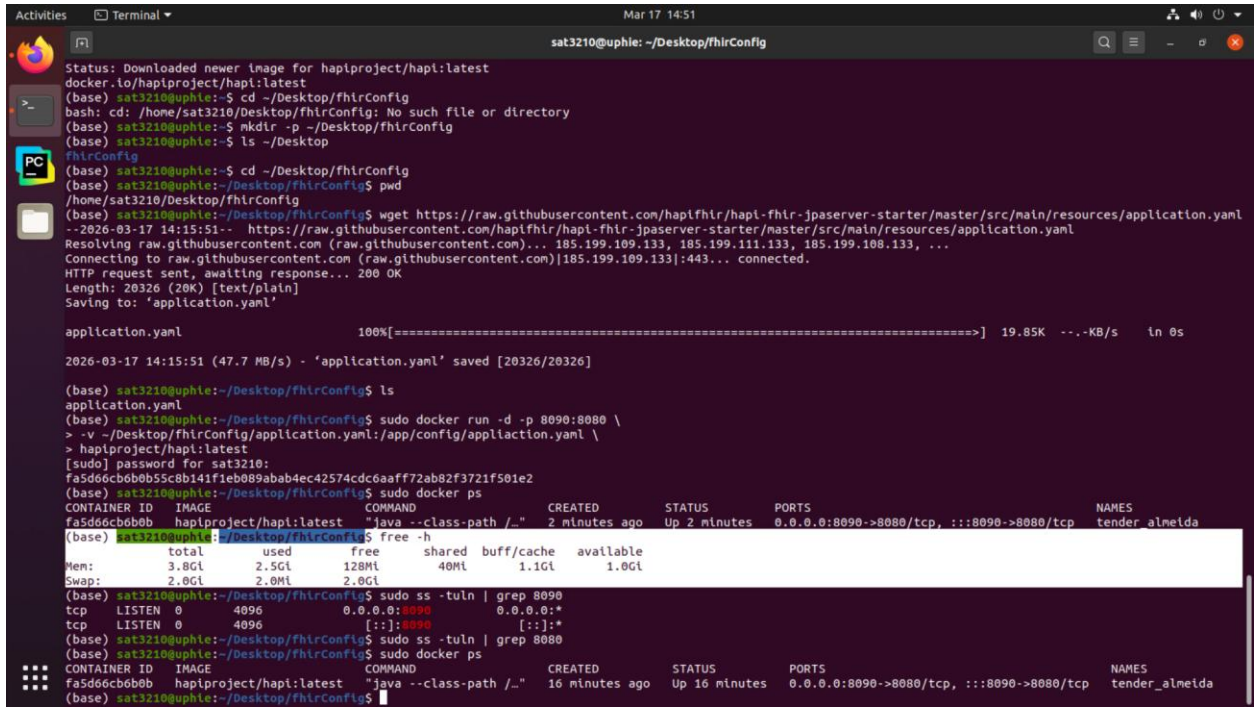
The screenshot shows the HAPI FHIR Server UI running in a Firefox Web Browser. The browser's address bar displays `localhost:8080`. A notification bar at the top of the browser indicates that it hasn't been started in a while and offers to refresh Firefox. The UI itself has a dark sidebar on the left with a 'Home' button and a list of resources: Account, ActivityDefinition, AdverseEvent, AllergyIntolerance, Appointment, AppointmentResponse, AuditEvent, and Basic. The main content area features the HAPI FHIR logo, a 'COMPANY NAME' placeholder, and a 'YOUR SAMPLE TEXT HERE' placeholder. Below this, there is a paragraph describing the server as a 100% open source implementation of the FHIR Specification. A table lists the server details:

Server	HAPI FHIR R4 Server
Software	HAPI FHIR Server - 8.8.0
FHIR Base	http://localhost:8080/fhir

Below the table, there is a 'Server Actions' section with a button labeled 'Conformance'.

2. Memory Check

I used the following command “**free -h**” as can be seen in the portion of the terminal I have highlighted in the screenshot below.



The screenshot shows a terminal window with the following commands and output:

```
Status: Download newer image for hapiproject/hapi:latest
docker.io/hapiproject/hapi:latest
(base) sat3210@uphie:~$ cd ~/Desktop/fhirConfig
bash: cd: /home/sat3210/Desktop/fhirConfig: No such file or directory
(base) sat3210@uphie:~$ mkdir -p ~/Desktop/fhirConfig
(base) sat3210@uphie:~$ ls ~/Desktop
fhirConfig
(base) sat3210@uphie:~$ cd ~/Desktop/fhirConfig
(base) sat3210@uphie:~/Desktop/fhirConfig$ pwd
/home/sat3210/Desktop/fhirConfig
(base) sat3210@uphie:~/Desktop/fhirConfig$ wget https://raw.githubusercontent.com/hapifhir/hapi-fhir-jpaserver-starter/master/src/main/resources/application.yaml
--2026-03-17 14:15:51-- https://raw.githubusercontent.com/hapifhir/hapi-fhir-jpaserver-starter/master/src/main/resources/application.yaml
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20326 (20K) [text/plain]
Saving to: 'application.yaml'

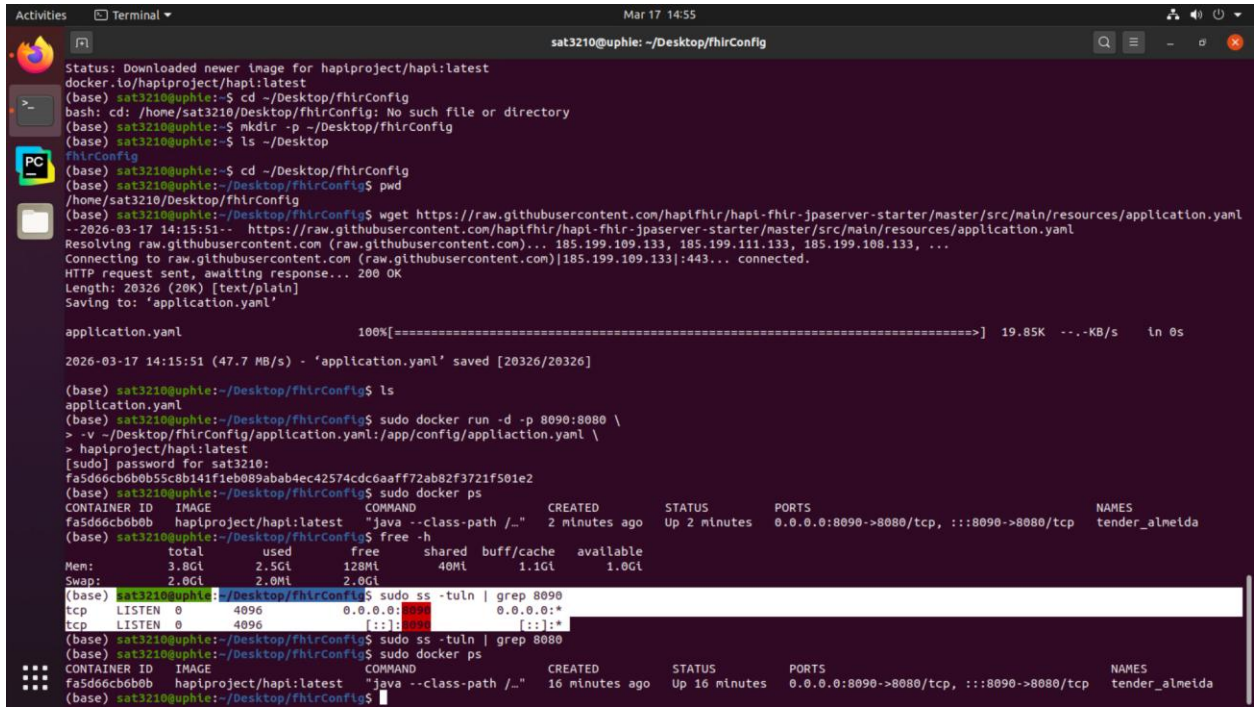
application.yaml      100%[=====] 19.85K  --.-KB/s  in 0s

2026-03-17 14:15:51 (47.7 MB/s) - 'application.yaml' saved [20326/20326]

(base) sat3210@uphie:~/Desktop/fhirConfig$ ls
application.yaml
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo docker run -d -p 8090:8080 \
> -v ~/Desktop/fhirConfig/application.yaml:/app/config/application.yaml \
> hapiproject/hapi:latest
[sudo] password for sat3210:
fa5d66cb6b0b hapiproject/hapi:latest "java --class-path /-"
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
fa5d66cb6b0b   hapiproject/hapi:latest "java --class-path /-"   2 minutes ago Up 2 minutes  0.0.0.0:8090->8080/tcp, :::8090->8080/tcp   tender_almeida
(base) sat3210@uphie:~/Desktop/fhirConfig$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.8Gi       2.5Gi       128Mi       40Mi        1.1Gi        1.0Gi
Swap:          2.0Gi       2.0Mi       2.0Gi
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo ss -tln | grep 8090
tcp        0         0          0.0.0.0:8090          0.0.0.0:*
tcp        0         0          [::]:8090          [::]:*
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo ss -tln | grep 8080
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
fa5d66cb6b0b   hapiproject/hapi:latest "java --class-path /-"   16 minutes ago Up 16 minutes  0.0.0.0:8090->8080/tcp, :::8090->8080/tcp   tender_almeida
(base) sat3210@uphie:~/Desktop/fhirConfig$
```

3. Port Availability Check

I used the following command “**sudo ss -tln | grep 8090**” as can be seen in the portion of the terminal I have highlighted in the screenshot below.



The screenshot shows a terminal window with the following content:

```
Activities Terminal Mar 17 14:55
sat3210@uphie: ~/Desktop/fhirConfig

Status: Download newer image for hapiproject/hapi:latest
docker.io/hapiproject/hapi:latest
(base) sat3210@uphie:~$ cd ~/Desktop/fhirConfig
bash: cd: /home/sat3210/Desktop/fhirConfig: No such file or directory
(base) sat3210@uphie:~$ mkdir -p ~/Desktop/fhirConfig
(base) sat3210@uphie:~$ ls ~/Desktop
fhirConfig
(base) sat3210@uphie:~$ cd ~/Desktop/fhirConfig
(base) sat3210@uphie:~/Desktop/fhirConfig$ pwd
/home/sat3210/Desktop/fhirConfig
(base) sat3210@uphie:~/Desktop/fhirConfig$ wget https://raw.githubusercontent.com/hapifhir/hapi-fhir-jpaserver-starter/master/src/main/resources/application.yaml
--2026-03-17 14:15:51-- https://raw.githubusercontent.com/hapifhir/hapi-fhir-jpaserver-starter/master/src/main/resources/application.yaml
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20326 (20K) [text/plain]
Saving to: 'application.yaml'

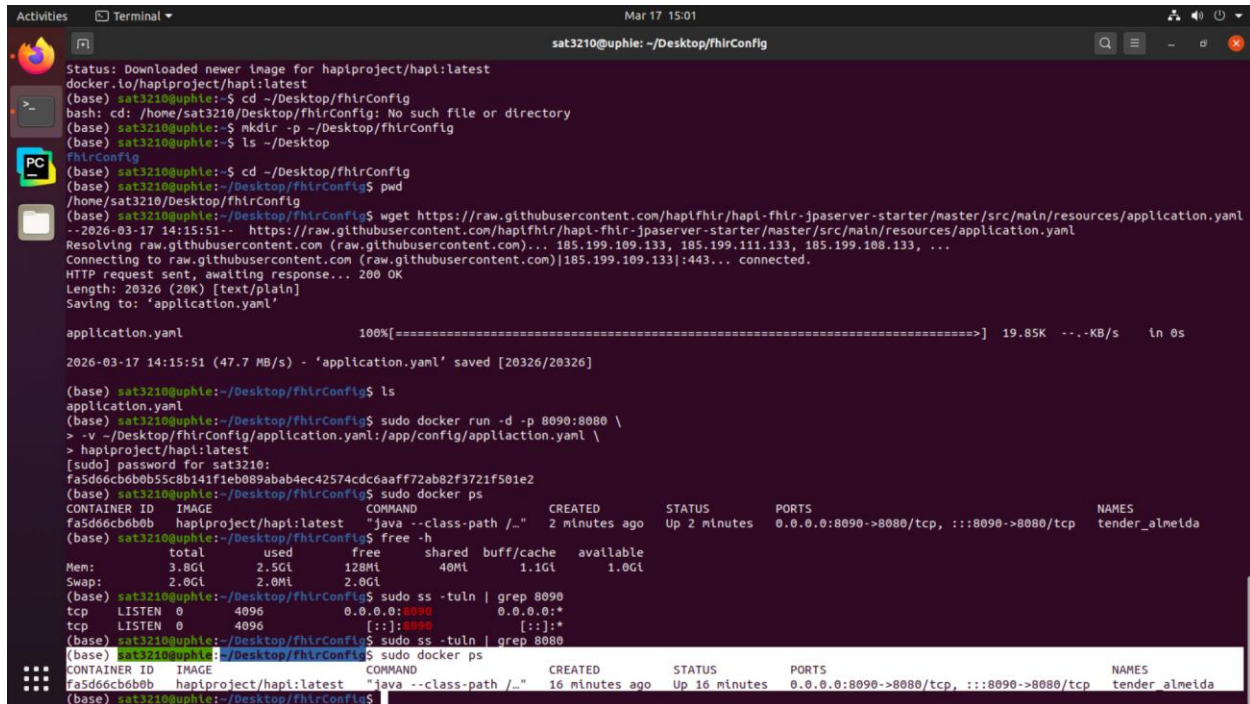
application.yaml      100%[=====] 19.85K  --.-KB/s  in 0s

2026-03-17 14:15:51 (47.7 MB/s) - 'application.yaml' saved [20326/20326]

(base) sat3210@uphie:~/Desktop/fhirConfig$ ls
application.yaml
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo docker run -d -p 8090:8080 \
> -v ~/Desktop/fhirConfig/application.yaml:/app/config/application.yaml \
> hapiproject/hapi:latest
[sudo] password for sat3210:
fa5d66cb6b0b hapiproject/hapi:latest "java --class-path ./-" 2 minutes ago Up 2 minutes 0.0.0.0:8090->8080/tcp, :::8090->8080/tcp tender_almeida
(base) sat3210@uphie:~/Desktop/fhirConfig$ free -h
             total        used        free      shared  buff/cache   available
Mem:           3.8Gi       2.5Gi       128Mi        40Mi        1.1Gi       1.0Gi
Swap:          2.0Gi       2.0Mi       2.0Gi
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo ss -tln | grep 8090
tcp        0          0          0.0.0.0:8090          0.0.0.0:*
tcp        0          0          [::]:8090          [::]:*
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo ss -tln | grep 8080
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo docker ps
CONTAINER ID   IMAGE               COMMAND                  CREATED        STATUS        PORTS                               NAMES
fa5d66cb6b0b  hapiproject/hapi:latest "java --class-path ./-" 16 minutes ago Up 16 minutes 0.0.0.0:8090->8080/tcp, :::8090->8080/tcp tender_almeida
(base) sat3210@uphie:~/Desktop/fhirConfig$
```

4. Port Usage Check (Port 8080)

I used the following command “**sudo docker ps**”. Port 8080 is used internally by the Docker container running the HAPI FHIR server. It is mapped to port 8090 on the host machine as can be seen in the portion of the terminal I have highlighted in the screenshot below.



The screenshot shows a terminal window with the following commands and output:

```
Status: Downloaded newer image for hapiproject/hapi:latest
docker.io/hapiproject/hapi:latest
(base) sat3210@uphie:~$ cd ~/Desktop/fhirConfig
bash: cd: /home/sat3210/Desktop/fhirConfig: No such file or directory
(base) sat3210@uphie:~$ mkdir -p ~/Desktop/fhirConfig
(base) sat3210@uphie:~$ ls ~/Desktop
fhirConfig
(base) sat3210@uphie:~$ cd ~/Desktop/fhirConfig
(base) sat3210@uphie:~/Desktop/fhirConfig$ pwd
/home/sat3210/Desktop/fhirConfig
(base) sat3210@uphie:~/Desktop/fhirConfig$ wget https://raw.githubusercontent.com/hapifhir/hapi-fhir-jpaserver-starter/master/src/main/resources/application.yaml
-2026-03-17 14:15:51- https://raw.githubusercontent.com/hapifhir/hapi-fhir-jpaserver-starter/master/src/main/resources/application.yaml
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.111.133, 185.199.108.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 20326 (20K) [text/plain]
Saving to: 'application.yaml'

application.yaml      100%[=====] 19.85K  --.-KB/s  in 0s

2026-03-17 14:15:51 (47.7 MB/s) - 'application.yaml' saved [20326/20326]

(base) sat3210@uphie:~/Desktop/fhirConfig$ ls
application.yaml
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo docker run -d -p 8090:8080 \
> -v ~/Desktop/fhirConfig/application.yaml:/app/config/application.yaml \
> hapiproject/hapi:latest
[sudo] password for sat3210:
fa5d66cb6b0b55c8b141f1eb089abab4ec42574cdc6aaff72ab82f3721f501e2
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS                               NAMES
fa5d66cb6b0b   hapiproject/hapi:late "java --class-path /..." 2 minutes ago  Up 2 minutes  0.0.0.0:8090->8080/tcp, :::8090->8080/tcp  tender_almeida
(base) sat3210@uphie:~/Desktop/fhirConfig$ free -h
              total        used        free      shared  buff/cache   available
Mem:           3.8Gi          2.5Gi        128Mi        40Mi         1.1Gi         1.0Gi
Swap:          2.0Gi          2.0Mi        2.0Gi
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo ss -tln | grep 8090
tcp        LISTEN     0          4096      0.0.0.0:8090  0.0.0.0:*
tcp        LISTEN     0          4096      [::]:8090    [::]:*
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo ss -tln | grep 8080
(base) sat3210@uphie:~/Desktop/fhirConfig$ sudo docker ps
CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS                               NAMES
fa5d66cb6b0b   hapiproject/hapi:late "java --class-path /..." 16 minutes ago  Up 16 minutes  0.0.0.0:8090->8080/tcp, :::8090->8080/tcp  tender_almeida
(base) sat3210@uphie:~/Desktop/fhirConfig$
```

Bonus Section

For the optional customization, I modified the HAPI FHIR server configuration in `application.yaml` by changing the tester/home branding values and adding UI title/subtitle settings. I then restarted the Docker container and verified from the logs that the application started successfully. In this container image, the homepage banner text remained hardcoded, but the server configuration itself was successfully customized and redeployed. I have attached and highlighted the section down below after I ran: **grep -n "Moses Nyirongo|hapi:|ui:|tester:" ~/Desktop/fhirConfig/application.yaml**

```
Activities Terminal Mar 19 23:35
sat3210@uphie: ~/Desktop/fhirConfig
2026-03-26T03:00:49.375Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling local job ca.uhn.fhir.jpa.cache.ResourceChangeListenerCacheRefresherImpl with interval 00:00:10.000
2026-03-26T03:00:49.379Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.jpa.search.reindex.ResourceReindexingServiceImpl with interval 00:00:10.000
2026-03-26T03:00:49.380Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.jpa.bulk.export.svc.BulkDataExportJobSchedulerImpl with interval 01:00:00
2026-03-26T03:00:49.383Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling local job ca.uhn.fhir.jpa.bulk.inprt.svc.BulkDataImportServiceImpl with interval 00:00:10.000
2026-03-26T03:00:49.390Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.jpa.search.warn.CacheWarningsServiceImpl with interval 00:00:10.000
2026-03-26T03:00:49.392Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.jpa.term.TermReadServiceImpl with interval 00:00:10.000
2026-03-26T03:00:49.393Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling local job ca.uhn.fhir.jpa.term.TermReindexingServiceImpl with interval 00:01:00.000
2026-03-26T03:00:49.394Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.jpa.search.SearchUrlJobMaintenanceServiceImpl with interval 00:10:00
2026-03-26T03:00:49.395Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.jpa.search.StateSearchDeletingServiceImpl with interval 00:01:00.000
2026-03-26T03:00:49.395Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling local job ca.uhn.fhir.jpa.util.ResourceCountCache with interval 00:10:00
2026-03-26T03:00:49.396Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling local job ca.uhn.fhir.jpa.subscription.triggering.SubscriptionTriggeringServiceImpl with interval 5000ms
2026-03-26T03:00:49.399Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.jpa.subscription.async.AsyncResourceModificationProcessingSchedulerSvc with interval 5000ms
2026-03-26T03:00:49.399Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.batch2.coordinator.ReductionStepExecutorServiceImpl with interval 00:00:10.000
2026-03-26T03:00:49.399Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Scheduling clustered job ca.uhn.fhir.batch2.maintenance.JobMaintenanceServiceImpl with interval 00:01:00.000
2026-03-26T03:00:49.400Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.u.f.j.sched.BaseSchedulerServiceImpl : Starting task schedulers for context hapi-fhir-jpaserver-starter
2026-03-26T03:00:49.400Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.uhn.fhir.jpa.sched.BaseHapiScheduler : Starting scheduler hapi-fhir-jpa-scheduler-local
2026-03-26T03:00:49.417Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] org.quartz.core.QuartzScheduler : Scheduler hapi-fhir-jpa-scheduler_$NON_CLUSTERED started.
2026-03-26T03:00:49.417Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] c.uhn.fhir.jpa.sched.BaseHapiScheduler : Starting scheduler hapi-fhir-jpa-scheduler-clustered
2026-03-26T03:00:49.417Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] org.quartz.core.QuartzScheduler : Scheduler hapi-fhir-jpa-scheduler_$NON_CLUSTERED started.
2026-03-26T03:00:49.488Z INFO 1 --- [hapi-fhir-jpaserver-starter] [main] ca.uhn.fhir.jpa.starter.Application : Started Application in 48.216 seconds (process running for 51.031)
(base) sat3210@uphie: ~/Desktop/fhirConfig$ nano ~/Desktop/fhirConfig/application.yaml
(base) sat3210@uphie: ~/Desktop/fhirConfig$ sudo docker restart tender_almeida
tender_almeida
(base) sat3210@uphie: ~/Desktop/fhirConfig$ sudo docker exec -it tender_almeida bash
OCI runtime exec failed: exec failed: unable to start container process: exec: "bash": executable file not found in $PATH: unknown
(base) sat3210@uphie: ~/Desktop/fhirConfig$ sudo docker exec -it tender_almeida sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "sh": executable file not found in $PATH: unknown
(base) sat3210@uphie: ~/Desktop/fhirConfig$ sudo docker exec tender_almeida grep -R "COMPANY NAME" -n /app /usr/local / 2>/dev/null | head
OCI runtime exec failed: exec failed: unable to start container process: exec: "grep": executable file not found in $PATH: unknown
(base) sat3210@uphie: ~/Desktop/fhirConfig$ grep -n "Moses Nyirongo|hapi:|ui:|tester:" ~/Desktop/fhirConfig/application.yaml
174:  name: Moses Nyirongo HAPI FHIR Server
196:  subtitle: Moses Nyirongo Bonus Customization
501:  hapi:
505:    ui:
507:      subtitle: Moses Nyirongo Bonus Customization
509:  tester:
(base) sat3210@uphie: ~/Desktop/fhirConfig$
```